

Question 1

You are given the following dataset of vehicles:

	Company	Model	Year	Price
0	Maruti	S-Cross	2013	1400000
1	Hyundai	Creta	2004	872000
2	MG	Hector	2016	1120000
3	Tata	Punch	2015	987650
4	Citreon	C3 Zeta	2019	560000

Write a Python program to:

- Create a DataFrame named 'Vehicle' from the given dataset.
- Reduce the price of each item by 50%.
- Change the values of the Year column to [2007, 2010, 2009, 2002, 2003].
- Draw a bar graph of Model vs Price with an appropriate title and axis labels.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

d1={'Company':['Maruti','Hyundai','MG','Tata','Citreon'],
    'Model':['S-Cross','Creta','Hector','Punch','C3 Zeta'],
    'Year':[2013,2004,2016,2015,2019],
    'Price':[1400000,872000,1120000,987650,560000]}
Vehicle=pd.DataFrame(d1)
print(Vehicle)

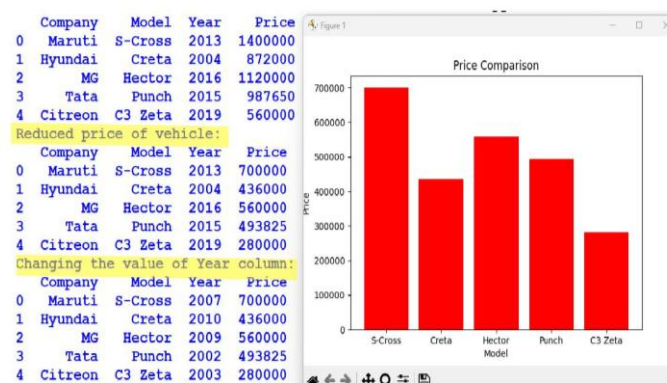
print("Reduced price of vehicle: ")
Vehicle['Price']=Vehicle['Price']//2          #ans of (ii)
print(Vehicle)

print("Changing the value of Year column: ")
Vehicle['Year']=[2007,2010,2009,2002,2003]   #ans of (iii)
print(Vehicle)

#Drawing bar graph on Model and Price; ans of (iv)
m=Vehicle['Model']
p=Vehicle['Price']
plt.title("Price Comparison")
plt.xlabel("Model")
plt.ylabel("Price")
plt.bar(m,p,color='r')
plt.show()
```

Output:

OUTPUT



Question 2

The monthly sales figures (in units) of three vehicle categories — Mid_Car, Bike and High_Car — over five months are given below. Write a Python program to store this data in a DataFrame and plot a vertical bar graph comparing the sales of all three categories month-wise, with a suitable title, axis labels and legend.

Code:

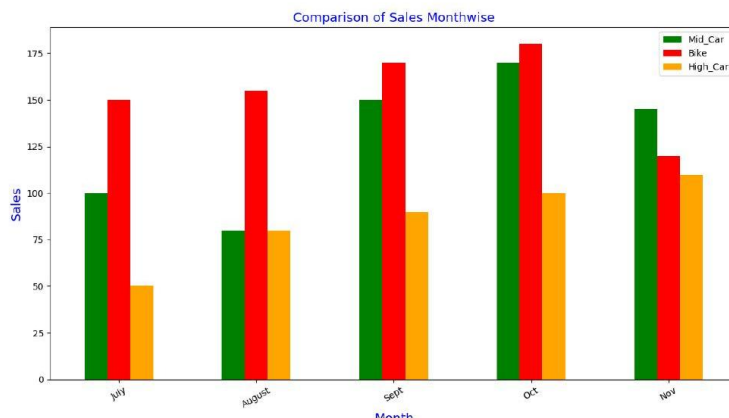
```
#Bar graph on multiple data items (vertical)
import matplotlib.pyplot as plt
import pandas as pd

dict1={'Mid_Car':[100,80,150,170,145],
      'Bike':[150,155,170,180,120],
      'High_Car':[50,80,90,100,110]}
df=pd.DataFrame(dict1,index=['July','August','Sept','Oct','Nov'])
print(df)

df.plot(kind='bar',color=['green','red','orange'])
plt.title("Comparison of Sales Monthwise",fontsize=14,color='blue')
plt.xlabel('Month',fontsize=14,color='blue')
plt.ylabel('Sales',fontsize=14,color='blue')
plt.xticks(fontsize=10,rotation=30)
plt.show()
```

Output:

	Mid_Car	Bike	High_Car
July	100	150	50
August	80	155	80
Sept	150	170	90
Oct	170	180	100
Nov	145	120	110



Question 3

Using the same sales dataset as Question 2, write a Python program to plot a horizontal bar graph (kind='barh') comparing the monthly sales of Mid_Car, Bike and High_Car, with a suitable title, axis labels and legend.

Code:

```
#Horizontal bar graph on multiple data items
import matplotlib.pyplot as plt
import pandas as pd

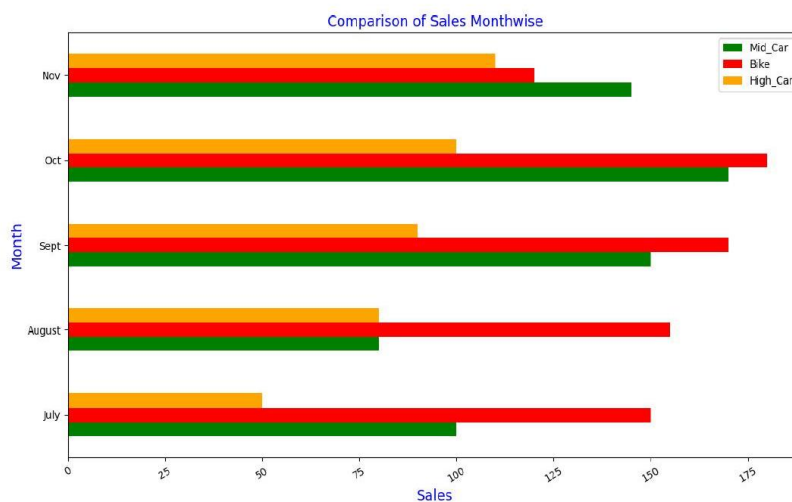
dict1={'Mid_Car':[100,80,150,170,145],
      'Bike':[150,155,170,180,120],
      'High_Car':[50,80,90,100,110]}
df=pd.DataFrame(dict1,index=['July','August','Sept','Oct','Nov'])
print(df)

df.plot(kind='barh',color=['green','red','orange'])
plt.title("Comparison of Sales Monthwise",fontsize=14,color='blue')
plt.ylabel('Month',fontsize=14,color='blue')
plt.xlabel('Sales',fontsize=14,color='blue')
plt.xticks(fontsize=10,rotation=30)
plt.show()
```

Output:

OUTPUT

	Mid_Car	Bike	High_Car
July	100	150	50
August	80	155	80
Sept	150	170	90
Oct	170	180	100
Nov	145	120	110



Question 4

The strength (number of students) of classes I to X in a school is given below:

Classes: I, II, III, IV, V, VI, VII, VIII, IX, X

Strength: 38, 30, 45, 49, 37, 53, 48, 44, 36, 46

Write a Python program to plot this data as a bar graph with an appropriate title, axis labels and a bar width of 0.82. Also save the chart as an image file named 'ch1.jpg'.

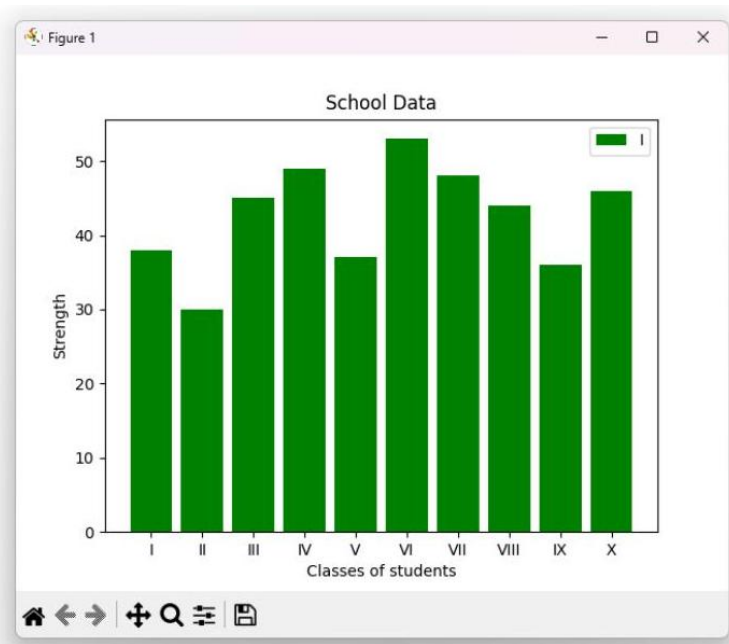
Code:

```
#Create a graph on strength of the students in classes I-X
import matplotlib.pyplot as plt

cl=['I','II','III','IV','V','VI','VII','VIII','IX','X']
st=[38,30,45,49,37,53,48,44,36,46]

plt.bar(cl,st,color='g',width=0.82)
plt.xlabel('Classes of students')
plt.ylabel('Strength')
plt.title('School Data')
plt.xticks(cl)
plt.legend(cl)
plt.savefig("ch1.jpg")
plt.show()
```

Output:



Question 5

The weekly sales figures (in Rs.) recorded over 7 days for three weeks (Week1, Week2, Week3) are given below. Write a Python program to:

- Store the data in a DataFrame and export it to a CSV file named 'sales.csv'.
- Read the CSV file back into a DataFrame and plot a vertical bar chart comparing Week1, Week2 and Week3 sales, with a title, axis labels and gridlines.

Code:

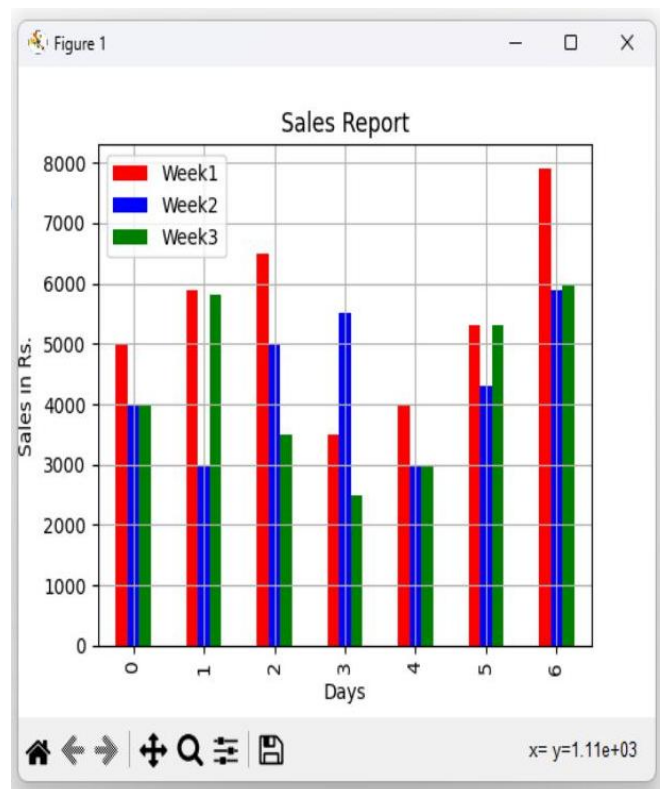
```
#Creating a bar chart from a CSV file
import pandas as pd
import csv
import matplotlib.pyplot as plt

df=pd.DataFrame(
    {'Week1':pd.Series([5000,5900,6500,3500,4000,5300,7900]),
     'Week2':pd.Series([4000,3000,5000,5500,3000,4300,5900]),
     'Week3':pd.Series([4000,5800,3500,2500,3000,5300,6000])})

export_csv=df.to_csv(r'd:\sales.csv',index=None,header=True)
print("Displaying the DataFrame: ")
print(df)

df=pd.read_csv("d:\sales.csv")
df.plot(kind='bar',color=['red','blue','green'])
plt.title("Sales Report")
plt.xlabel("Days")
plt.ylabel("Sales in Rs.")
plt.grid(True)
plt.show()
```

Output:



Question 6

Using the same weekly sales CSV data as Question 5, write a Python program to plot a horizontal bar chart (kind='barh') comparing Week1, Week2 and Week3 sales, with a title, axis labels and gridlines.

Code:

```
#Creating a horizontal bar chart from a CSV file
import pandas as pd
import csv
import matplotlib.pyplot as plt

df=pd.DataFrame(
    {'Week1':pd.Series([5000,5900,6500,3500,4000,5300,7900]),
     'Week2':pd.Series([4000,3000,5000,5500,3000,4300,5900]),
     'Week3':pd.Series([4000,5800,3500,2500,3000,5300,6000])})

export_csv=df.to_csv(r'd:\sales.csv',index=None,header=True)
print("Displaying the DataFrame: ")
print(df)

df=pd.read_csv("d:\sales.csv")
df.plot(kind='barh',color=['red','blue','green'])
plt.title("Sales Report")
plt.ylabel("Days")
plt.xlabel("Sales in Rs.")
plt.grid(True)
plt.show()
```

Output:



Question 7

You are provided with a structured dataset containing the academic results of Class XII students across different streams. Write a Python program using the pandas and matplotlib libraries to perform the following operations:

- Create a DataFrame containing student data (Roll No, Name, Section, Stream, and marks in 4 subjects).
- Calculate the 'Total Marks' and 'Percentage' for each student and add them as new columns.
- Analyse the data by calculating the average percentage scored by each Section.
- Generate a bar chart to visualise the Section-wise average percentage. Ensure the chart includes an appropriate title, X and Y axis labels, and a legend.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Creating the DataFrame mimicking structured result data
data = {
    'RollNo': [2201, 2202, 2203, 2204, 2205, 2206, 2207, 2208],
    'Name': ['Aarav', 'Bhavya', 'Chirag', 'Diya', 'Eshaan', 'Fiza', 'Gaurav', 'Hina'],
    'Section': ['XII-A', 'XII-A', 'XII-A', 'XII-B', 'XII-B', 'XII-C', 'XII-C', 'XII-C'],
    'Stream': ['Science', 'Science', 'Science', 'Commerce', 'Commerce', 'Commerce', 'Humanities', 'Humanities', 'Humanities'],
    'English': [88, 92, 75, 85, 78, 90, 85, 92],
    'Sub_2': [95, 89, 80, 90, 88, 85, 82, 88],
    'Sub_3': [92, 94, 78, 85, 80, 88, 79, 91],
    'Sub_4': [98, 90, 85, 95, 85, 92, 88, 89]
}
df = pd.DataFrame(data)
print("--- Original Result Data ---")
print(df)

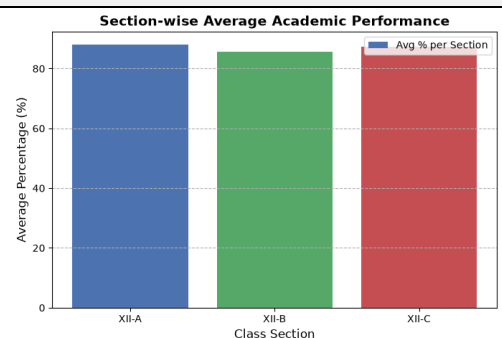
# 2. Calculating Total Marks and Percentage
df['Total'] = df['English'] + df['Sub_2'] + df['Sub_3'] + df['Sub_4']
df['Percentage'] = (df['Total'] / 400) * 100
print("--- Data with Total and Percentage ---")
print(df[['Name', 'Section', 'Total', 'Percentage']])

# 3. Section-wise average percentage
section_analysis = df.groupby('Section')['Percentage'].mean().reset_index()
print("--- Section-wise Average Percentage ---")
print(section_analysis)

# 4. Plotting the Bar Chart
plt.figure(figsize=(8, 5))
sections = section_analysis['Section']
averages = section_analysis['Percentage']
plt.bar(sections, averages,
        color=['blue', 'green', 'red'], label='Avg % per Section')
plt.title('Section-wise Average Academic Performance', fontsize=14, fontweight='bold')
plt.xlabel('Class Section', fontsize=12)
plt.ylabel('Average Percentage (%)', fontsize=12)
plt.legend()
plt.grid(axis='y', linestyle='--')
plt.show()
```

Output:

```
--- Section-wise Average Percentage ---
  Section  Percentage
0  XII-A      88.000000
1  XII-B      85.750000
2  XII-C      87.416667
```



Question 8

You have downloaded a dataset from an open-source portal (e.g., data.gov.in) detailing the annual renewable energy capacity (in Megawatts) installed across different Indian states over three years. Write a Python program to:

- (i) Load this data into a pandas DataFrame.
- (ii) Aggregate the data to find the total renewable energy capacity installed by each state over the given years.
- (iii) Summarise the year-wise total national capacity.
- (iv) Plot a bar chart comparing the state-wise total renewable energy capacity, with an appropriate title, axis labels and distinct bar colours.
- (v) Plot a line chart showing the year-wise growth in national capacity, with markers, a title and a legend.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt

# 1. Creating DataFrame mimicking open-source government data
gov_data = {
    'State': ['Gujarat', 'Tamil Nadu', 'Karnataka', 'Rajasthan', 'Maharashtra',
              'Gujarat', 'Tamil Nadu', 'Karnataka', 'Rajasthan', 'Maharashtra',
              'Gujarat', 'Tamil Nadu', 'Karnataka', 'Rajasthan', 'Maharashtra'],
    'Year': [2021, 2021, 2021, 2021, 2021, 2022, 2022, 2022, 2022, 2022,
             2023, 2023, 2023, 2023, 2023],
    'Capacity_MW': [1500, 1800, 2000, 2200, 1300, 1700, 1950, 2300, 2600, 1450,
                    2100, 2200, 2700, 3100, 1800]
}
df_energy = pd.DataFrame(gov_data)

# 2. State-wise total capacity
state_total = df_energy.groupby('State')['Capacity_MW'].sum()
print("--- State-wise Total Renewable Capacity (MW) ---")
print(state_total)

# 3. Year-wise total national capacity
year_total = df_energy.groupby('Year')['Capacity_MW'].sum()
print("--- Year-wise National Capacity (MW) ---")
print(year_total)

# 4 & 5. Plotting the charts side by side
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 6))

# Bar chart: state-wise contribution
ax1.bar(state_total.index, state_total.values)
ax1.set_title('Total Renewable Capacity by State (2021-2023)', fontweight='bold')
ax1.set_xlabel('State')
ax1.set_ylabel('Total Capacity (MW)')
ax1.grid(axis='y', linestyle='--')

# Line chart: year-wise national growth
ax2.plot(year_total.index, year_total.values, marker='o', color='green',
         linestyle='-', linewidth=2, markersize=8, label='Total MW')
ax2.set_title('National Renewable Energy Growth', fontweight='bold')
ax2.set_xlabel('Year')
ax2.set_ylabel('Capacity Installed (MW)')
ax2.set_xticks([2021, 2022, 2023])
ax2.grid(True, linestyle=':')
ax2.legend()

plt.tight_layout()
plt.show()
```


Output:

--- State-wise Total Renewable Capacity (MW) ---	
State	
Gujarat	5300
Karnataka	7000
Maharashtra	4550
Rajasthan	7900
Tamil Nadu	5950
--- Year-wise National Capacity (MW) ---	
Year	
2021	8800
2022	10000
2023	11900

